Which origin countries drive the effect of immigration on employment? Evidence from Australia *

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Abstract

Shift-share instruments are widely used to estimate the labor market effects of immigration, yet the variation that drives them is rarely determined. This paper studies Australia, where immigrants make up nearly 30 percent of the population, using census data. We instrument the immigrant share with the 1991 settlement patterns of 29 origin groups and decompose the estimates using Rotemberg weights. The identifying variation is concentrated in India, China, the Philippines, Vietnam, and Malaysia account for more than half of the total weight, and negative weights are negligible. Regions that received larger immigrant inflows experienced higher employment rates, and a shift in employment from middle-wage toward high-wage occupations. This finding suggests that Asian immigration has prompted changes in the occupational composition of Australian local labor markets.

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1 Introduction

Like the United States, Australia has long been a country of immigrants. As of 2025, approximately 32% of the Australian population was born overseas (Australian Bureau of Statistics, 2025a). Until the early 1970s, under the White Australia Policy, Australia admitted immigrants primarily from Britain and other European countries. The policy was officially abolished in 1973, and the share of Asian immigrants gradually increased with the resettlement of Vietnamese refugees in the late 1970s. However, these immigrants arrived largely as refugees or through family reunion rather than on the basis of their skills. After 1996, the government shifted its immigration policy from family-based to skill-based immigration policy (Larsen, 2013). Since then, the inflow of Asian immigrants admitted through skill-based migration and labor market demand has increased substantially. This shift raises the question of how these immigrants affect local labor markets and which origin countries drive these effects.

In this paper, we use Rotemberg weights to identify the origin countries that drive the effects of immigration on local labor markets in Australia. We examine employment, labor force participation, unemployment, and occupational composition across local government areas (LGAs) from 2001 to 2021. To address the endogeneity of immigrant location choices, we use a shift-share instrument that allocates national-level immigrant inflows to LGAs based on historical settlement patterns of immigrants from each origin country prior to Australia's immigration policy reform in 1996.

Estimating the effects of immigration on local labor markets is complicated by the fact that immigrants choose where to settle, and are likely to be drawn to areas with favorable economic conditions. Following Card (2001), we address this concern with a shift-share instrument that combines the national stock of immigrants from each origin country (the "shift") with the 1991 distribution of immigrants from that country across LGAs (the "share"). The instrument builds on the observation that immigrants tend to settle where earlier immigrants from the same origin live, so that historical enclaves predict subsequent inflows while being plausibly unrelated to contemporaneous local economic shocks.

Recent work has re-examined the conditions under which such instruments are valid. Goldsmith-Pinkham et al. (2020) show that the shift-share estimator is equivalent to a GMM estimator that uses the initial shares as instruments, implying that identification rests on the exogeneity of the shares, whereas Borusyak et al. (2022) show that identification can instead come from many exogenous shocks. Jaeger et al. (2018) further caution that, when the shares and shifts are highly serially correlated, shift-share instruments may conflate the short-run effects of new inflows with the long-run adjustment to past ones. These developments imply that researchers should make explicit which identifying assumption their design relies on and which sources of variation drive their estimates.

We contribute to this discussion by making the identifying variation of our instru-

ment transparent. Since our instrument reflects the differential exposure of LGAs to national immigration growth through their 1991 enclaves, we adopt the share-based view of Goldsmith-Pinkham et al. (2020) and decompose our estimates using their Rotemberg weights. This decomposition shows that India and China together account for about 45% of the identifying variation and the top five origin countries for 57%, that negative weights play a negligible role, and that the just-identified estimates of the highest-weighted countries are consistent in sign and magnitude with our main estimates. We further show that our results are robust to leave-one-country-out instruments that exclude India or China, and we discuss the extent to which the persistence of national immigrant stocks raises the concern of Jaeger et al. (2018) in our setting.

The remainder of the paper is organized as follows. Section **??** describes the data and variable construction. Section **??** presents the empirical strategy. Section 2.1 and Section 2.2 describe the FE-IV model and the shift-share instrument, and Section 2.3 describes the Rotemberg weight decomposition. Section 3 reports the main results on labor force outcomes and occupational composition. Section 4 reports robustness checks using leave-out instruments, and Section 5 concludes.

2 Data and method

We use census-based community profiles at the local government area (LGA) level, collected at a quinquennial frequency by the Australian Bureau of Statistics (ABS). These profiles are constructed using the full census enumeration and include information on the number of foreign-born residents, the number of native-born residents, and a range of LGA-level labor market and demographic outcomes. Beginning in 2001, the ABS started releasing time-series community profiles that compile information across three consecutive census waves.

For example, the 2001 time-series community profiles provide LGA-level data for the census years 1991, 1996, and 2001. We select profile releases to maximize consistency within each variable type. Immigrant counts by country of origin, which are required for IV construction, are drawn from the 2001, 2016, and 2021 time-series profiles, as these provide harmonized country of origin classifications spanning 1991 to 2021. Labor market outcomes are drawn from the 2006 and 2021 time-series profiles, which provide consistently defined employment and demographic variables. Our analysis examines the effects of immigration on local labor market outcomes including employment-to-population ratio, the share of population not in the labor force, and the occupational composition of employment. We construct the immigrant share using 1991 as the base-year population denominator, defined as the total LGA population in 1991. This choice is motivated by the structure of our shift-share instrumental variable, which uses 1991 as the base year for the initial settlement share. Following the shift-share literature (e.g., Card (2001);

Peri (2012)), we allow for a lag between the base year and the start of the estimation sample to ensure that the initial settlement patterns predate the period of analysis and are thus less susceptible to concerns about reverse causality. The final estimation sample therefore consists of a balanced panel of 496 LGAs observed from 2001 to 2021.

To construct the key explanatory variable, the immigration share, we define immigrants as individuals born outside of Australia territory, identified through birthplace information from the ABS data. Across the entire sample period, we consolidate all reported birthplaces into 29 origin groups to ensure comparability over time because the set of birthplaces recorded in the census varies across waves. These groups consist of 28 major countries and one residual category that aggregates all remaining, less consistently observed birthplaces. We then define the immigrant share in a county as the number of immigrants in the current year divided by the countys total population in 1991.

Our outcome variables comprise two sets of measures, all constructed at the LGA level from raw census counts. First, labor force status measures are constructed for the population aged 15 and above. The employment share is defined as the number of employed individuals divided by the population aged 15 and above, and the share not in the labor force is defined analogously. The unemployment rate is defined as the number of unemployed individuals divided by the labor force. We construct each measure for the overall population and separately for males and females, where the denominators are the corresponding male and female populations. Second, occupational employment shares are defined as the number of employed individuals in each occupational group divided by the total number of employed individuals. We construct both overall and gender-specific measures, where the denominator is the total number of male and female employed individuals, respectively. We group the eight ANZSCO major groups into three tiers based on median hourly earnings reported by the Australian Bureau of Statistics (2025b). The high-wage tier comprises managers and professionals, whose median hourly earnings (about $50 in 2020) are well above those of all other groups. The middle-wage occupation comprises clerical and administrative workers, technicians and trades workers, machinery operators and drivers, and community and personal service workers (about $30 to $34). The low-wage occupation comprises labourers and sales workers (about $26).

Table 1 reports the descriptive statistics for the main estimation sample, comprising 496 LGAs observed over five census year from 2001 to 2021, for a total 2,480 observations. The immigration share averages 27.7% across LGAs, with a standard deviation of 22.7%, reflecting considerable heterogeneity in immigrant concentration across local areas. On average, 56.3% of the population aged 15 and above is employed and 33.6% is not in the labor force, and the mean unemployment rate is 5.6%. Turning to the occupational composition of employment, middle-wage occupations account for the largest share of employment (42.4%), followed by high-wage occupations (35.4%) and low-wage occupations (20.2%).

[Table 1]

2.1 Model

To examine the effects of immigration on local labor market outcomes, we consider the following model:

$$y_{it} = \beta ImmiShare_{it} + (Z_{i,0} \times t)'\gamma + \rho_i + \phi_t + \varepsilon_{it} \tag{1}$$

where the dependent variable, y_{it}, denotes an employment or occupational employment share in LGA i in year t. The main explanatory variable, $ImmiShare_{it}$, is the share of immigrants in LGA i in year t. $Z_{i,0}$ is a vector of pre-sample LGA characteristics, the population share aged 65 and above and the college graduate share, measured five years before the start of the sample period, and t is a linear time trend. These interactions allow LGAs with different initial demographic structures and levels of human capital to follow different trends in labor market outcomes. Because the characteristics are measured before the sample period, they are not affected by subsequent immigration. LGA fixed effects ρ_i absorb time-invariant local characteristics, such as geographic and institutional factors, while census-year fixed effects ϕ_t control for aggregate macroeconomic shocks common to all LGAs in a given period, and ε_{it} is an idiosyncratic error term.

The coefficient of interest, β, measures the impact on each outcome variable of a one percentage-point rise in the immigration share. Whether β in equation (1) admits a causal interpretation depends critically on the exogeneity of local immigration share variation with respect to labor market outcomes, given the controls included in the specification. We proceed by first outlining how the immigration share measure is constructed, before turning to the identification strategy we employ to isolate the causal effect of immigration exposure on local labor market outcomes.

2.2 Identification strategy

We construct the immigration share, $ImmiShare_{it}$, as the total number of immigrants residing in LGA i in year t, divided by the total population of the LGA in 1991. Letting M_{ikt} denote the number of immigrants from origin country k residing in LGA i in year t, we can write $ImmiShare_{it}$ as:

$$ImmiShare_{it} = \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} M_{ikt} = \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} \frac{M_{ikt}}{M_{kt}} M_{kt}, \tag{2}$$

As shown in equation (2), the immigration share decomposes into two components, the national-level stock of immigrants from origin country k in year t (i.e., M_{kt}, the shift component), and the share of those national immigrants living in LGA i (i.e., $\frac{M_{ikt}}{M_{kt}}$, the

share component). Identifying the causal effect of $ImmiShare_{it}$ on local outcomes requires that its variation be exogenous to unobserved factors driving changes in employment and occupational composition at the LGA level. However, this assumption is unlikely to hold in practice, as the two components of equation (2) raise distinct endogeneity concerns.

First, the shift component, M_{kt}, is primarily determined by aggregate push factors in origin countries and by national immigration policies, rather than by local economic conditions in any single LGA. Consequently, changes in employment and occupational composition in a given LGA are unlikely to influence origin-specific national immigration stocks. For this reason, M_{kt} can be regarded as exogenous to local labor market conditions at the LGA level.

However, the second component, $\frac{M_{ikt}}{M_{kt}}$, is potentially problematic. Changes in local job opportunities within in an LGA may influence the composition of immigrant groups who choose to settle there, giving rise to endogeneity in this component. While LGA fixed effects in equation (1) partially mitigate this concern by absorbing time-invariant local characteristics, the presence of transitory local shocks remains a source of bias. Specifically, short-run economic shocks or changes in local amenities may simultaneously affect local outcomes and the sorting of immigrants across LGAs.

To address this endogeneity concern, we replace the contemporaneous share component with its predetermined counterpart from 1991, prior to the start of our sample period, yielding a shift-share instrument of the following form:

$$ImmiShare_{it}^{IV} \equiv \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} \frac{M_{ik,1991}}{M_{k,1991}} M_{kt}, \tag{3}$$

This instrument takes the form of a weighted sum across origin countries, where the weights are given by the predetermined settlement shares, $\frac{M_{ik,1991}}{M_{k,1991}}$, defined as the share of origin country k immigrants residing in LGA i in the base year 1991. These settlement shares are then applied to the national-level stock of immigrants from country k in year t, M_{kt}, to generate a predicted immigration share for each LGA-year cell. Intuitively, $ImmiShare_{it}^{IV}$ captures the immigrant population that LGA i would be expected to have in year t based solely on its historical settlement patterns, thereby isolating variation in the local immigration share that is driven by national-level changes in immigration rather than by contemporaneous local conditions.

We select 1991 as the base year for the settlement shares for several reasons following Kim and Wang (2024). First, it is the earliest census year for which detailed country of origin information is publicly available in the ABS data. Second, it predates the reorientation of the migration program from family-based to skill-based admission in 1996, after which skill-based visas became the dominant pathway to permanent residence and the immigrant share began to rise rapidly (**?**). Settlement shares measured in 1991 therefore

reflect location choices made before this structural shift in the scale and composition of immigration. Third, immigrant settlement patterns in 1991 predate the pronounced changes in the country of origin composition of immigrants documented since the 1990s (Miller, 1999). As shown in Figure 1, the share of immigrants from Asia roughly doubled between 1991 and 2021, from about 15% to 30%, while the share from Europe fell from 44% to 20%, with the Asian share overtaking the European share in the mid-2010s.

[Figure 1]

To establish the relevance of this instrument, consider each component of $ImmiShare_{it}^{IV}$ in equation (3) and compare it with the right-hand side of equation (2). First, both share the same origin-specific national immigrant stocks in year t, M_{kt}, which can be regarded as exogenous for the reasons outlined above. Second, rather than relying on contemporaneous settlement shares, $\frac{M_{ikt}}{M_{kt}}$, the instrument exploits the predetermined settlement share of each immigrant group, $\frac{M_{ik,1991}}{M_{k,1991}}$, within an LGA. The predictive power of historical settlement patterns for subsequent location choices stems from a well-established tendency for immigrants to concentrate in areas that already host sizable communities from the same country of origin (Bartel, 1989).

The validity of the instrument requires that ε_{it} in equation (1) be uncorrelated with the factors that shaped the settlement choices of immigrants in 1991. We argue that this condition is satisfied for two reasons. First, local economic or demographic conditions within a single LGA are unlikely to drive national-level immigration trends, which rules out any direct feedback from local outcomes to the shift component, M_{kt}. Second, the substantial time elapsed between the base year settlement patterns and subsequent changes in local outcomes, together with the conditioning on LGA and year fixed effects and on pre-sample LGA characteristics interacted with a linear time trend in equation (1), renders it implausible that ε_{it} is systematically related to the geographic distribution of immigrants in 1991. Because identification ultimately rests on the exogeneity of the 1991 settlement shares (Goldsmith-Pinkham et al., 2020), we use Rotemberg weights to identify the origin groups that drive the identifying variation and show that the results are robust to excluding the most influential origins, India and China, from the instrument (Section 4).

2.3 Rotemberg weights

A growing literature has revisited the validity of shift-share instruments (Jaeger et al., 2018; Borusyak et al., 2022). Goldsmith-Pinkham et al. (2020) show that the Bartik estimator is numercially equivalent to a GMM estimator that uses the initial shares as instruments, so that identification rests on the exogeneity of the shares. Borusyak et al. (2022) instead derive identification from many independent shocks. Our instrument

reflects differential exposure of LGAs to common national immigration growth through their 1991 enclaves, which Goldsmith-Pinkham et al. (2020) associate with share-based identification. Moreover, with only 29 origin groups and highly persistent national stocks (Jaeger et al., 2018), the many-shocks argument is less plausible in our setting.

Table 2 shows that the top five instruments account for about 55 percent (0.558/1.026)of positive weight in the estimator. In Panel B, the correlation between $\hat{\alpha}_k$ and the shift g_k is 0.582, so the shifts explain about one third of the variance in the weights. In contrast, the correlation with the cross-LGA variance of the shares is -0.171, indicating that countries receive large weights mainly because their national stocks grew rapidly, consistent with their small 1991 shares of 1.5–3.0%. The weights are positively correlated with the first-stage F-statistics (0.314), so that the countries contributing most to the identifying variation also tend to be stronger individual instruments. Finally, the correlation between the weights and the just-identified estimates is weak (-0.127), suggesting that the highest-weighted countries do not systematically yield estimates that diverge from the overall estimate.

[Table 2]

Panel C reports the five origin countries with the largest Rotemberg weights. India (28.1%) and China (16.6%) receive the largest weights, and the top five countries together account for 57.3% of the total weight. Following the logic of Goldsmith-Pinkham et al. (2020), our estimates are thus driven mainly by comparisons between LGAs with more and fewer Indian and Chinese immigrants in 1991, and the key concern is whether these LGAs have other characteristics that predict changes in labor market outcomes through non-immigration channels. Reassuringly, the just-identified estimates of all five countries are positive and statistically significant, with India (0.320) and China (0.299) close to the main estimate of 0.330.

Panel D splits the main estimate into the α-weighted sums of the just-identified estimates for countries with positive and negative weights, which add up to the main estimate. If the weighted sum for countries with negative weights were large relative to that for countries with positive weights, negatively weighted estimates would be driving the overall estimate. While negative weights pose no problem under constant effects, under heterogeneous effects they could undermine the interpretation of the estimate as a weighted average of origin-specific effects. In our setting, countries with positive weights contribute 0.353, while those with negative weights contribute only -0.023, about 6.5% of the former. This ratio is similar to the roughly 6% in the canonical setting of Goldsmith-Pinkham et al. (2020) and well below the 12–23% found by Wright (2026) for Hunt and Gauthier-Loiselle (2010), and it is below 3% for the other outcomes (Table A1, Table A2, and Table A3). Negative weights therefore play a negligible role in our estimates.

3 Results

Table 4 presents the estimated effects of the immigration share on labor force outcomes using Pooled OLS, FE. The dependent variables are employment share, the share of population not in the labor force, and the unemployment rate, reported in Columns (1) through (3), respectively. Panel A reports pooled OLS estimates, controlling for year fixed effects and the control variables. The immigration share is positively and significantly associated with the unemployment rate, weakly negatively associated with the share not in the labor force, and not significantly associated with the employment share. Panel B reports FE estimates, which add LGA fixed effects. The immigration share is positively and significantly associated with the employment share and negatively with the share not in the labor force, while the positive association with the unemployment rate in Panel A disappears.

[Table 4]

Panel A of Table 5 presents the FE-IV estimates of the effects of the immigration share on labor force outcomes. The diagnostics indicate that the instrument is strong. The first-stage coefficient is 0.807, and the Kleibergen-Paap F-statistic is approximately 23. We also report the Anderson-Rubin F-statistics and p-values. The FE-IV estimates show that the immigration share significantly increases employment share, with a coefficient of 0.330. This effect is not detected in the Pooled OLS estimate and is much smaller in FE estimate. The share not in the labor force shows a similar pattern. The coefficient is negative in all three specifications, but its magnitude grows from pooled OLS to FE-IV, which suggests that both the pooled OLS and FE estimates understate the effect. In contrast, the unemployment rate is positive and significant in pooled OLS (0.014), but becomes small and insignificant once LGA fixed effects are included and remains insignificant in FE-IV estimation. The coefficient is positive and significant in pooled OLS (0.014), but becomes small and insignificant once LGA fixed effects are included, and remains insignificant in FE-IV (-0.026). This suggests that immigrants tend to live in LGAs with persistently higher unemployment rather than that immigration raises unemployment.

[Table 5]

We now turn to the magnitudes of the FE-IV estimates. A one standard deviation increase in the immigration share, approximately 22.7 percentage points, raises the employment share by 7.5 percentage points relative to a mean of 56.3%. It lowers the share not in the labor force by 3.0 percentage points relative to a mean of 33.6%. Panel B and C of Table 5 show that these patterns hold for both men and women. For men, a one standard deviation increase in the immigration share raises the employment share by 8.6

percentage points. For women, the employment share also rises by 6.0 percentage points and the share not in the labor force falls by 1.5 percentage points. For both men and women, the effect on the unemployment rate is insignificant.

To provide a more nuanced understanding of the employment effects documented in Table 5, we examine whether immigration changes the occupational composition of employment. Table 6 reports FE-IV estimates for the shares of high-, middle-, and low-wage occupations among the total employed. Panel A reports the results for all workers, and Panel B and C report the results for men and women.

[Table 6]

Panel A shows that immigration shifts employment toward high-wage occupations. A one standard deviation increase in the immigration share raises the share of high-wage occupations by 6.9 percentage points, relative to a mean of 35.4%. It lowers the share of middle-wage occupations by 6.2 percentage points, relative to a mean of 42.4%. The effect on the share of low-wage occupations is small and insignificant. Panels B and C show that this pattern holds for both men and women. For men, the share of high-wage occupations rises by 6.4 percentage points and the share of middle-wage occupations falls by 6.2 percentage points. For women, the share of high-wage occupations rises by 7.9 percentage points and the share of middle-wage occupations falls by 6.1 percentage points. For both men and women, the effect on the share of low-wage occupations is insignificant.

4 Robustness checks

Robustness check - (leave out IV) We examine the robustness of our main results on labor market outcomes to the concentration of the identifying variation in a small number of origin countries. As discussed in Section 2.2, the validity of the shift-share IV rests on the exclusion restriction that unobserved factors influencing local labor market conditions are independent of the historical settlement patterns of immigrants across LGAs in 1991. Recent research has raised concerns about identification with shift-share IVs, particularly when the initial shares used as weights may themselves be endogenous (Jaeger et al., 2018; Goldsmith-Pinkham et al., 2020). As shown in Section 2.3, India and China receive the largest Rotemberg weights (0.281 and 0.166, respectively), together accounting for about 45% of the total weight. If the settlement shares of either group were correlated with unobserved local shocks, our estimates could be biased. To assess whether our main findings are sensitive to this concentration, we construct alternative shift-share instruments that exclude immigrants from India or from China (Kim and Wang, 2024).

leave out IV Table 7 reports the results for labor force status. Excluding India or China from the instrument yields estimates that are nearly identical to those in Table 5.

The effect on the employment share ratio is 0.334 when India is excluded and 0.336 when China is excluded, compared with 0.330 in the main specification, and the effect on the share not in the labor force remains negative and significant. The effect on the unemployment rate remains small and statistically insignificant. Table 8 reports the results for occupational composition. The effect on the high-wage occupation share is 0.324 and 0.320, compared with 0.306 in the main specification, and the effect on the middle-wage share remains negative and significant, while the effect on the low-wage share remains close to zero. The first-stage Kleibergen-Paap F-statistics are 15.8 and 23.7, respectively. These results confirm that our findings are not driven by immigrants from India or China alone.

[Table 7]

Table 8 reports the corresponding results for occupational composition. When India is excluded from the instrument, the estimated effect on the high-wage occupation share is 0.324, and the effect on the middle-wage occupation share is -0.306, both statistically significant at the 1% level. When China is excluded, the corresponding estimates are 0.320 and -0.279, also significant at the 1% level. These estimates are close to those in Table 6 (0.306 and -0.275, respectively). The effect on the low-wage occupation share remains small and statistically insignificant in both specifications (0.006 and -0.016). The first-stage Kleibergen-Paap F-statistics are 15.8 and 23.7 when India and China are excluded, respectively.

[Table 8]

5 Conclusion

1st paragraph : summary (methodology) This paper investigates the causal effects of immigration on local labor market outcomes in Australia, using census data from the Australian Bureau of Statistics (ABS) across LGAs. To address the endogeneity of the immigration share, we employ a fixed effects instrumental variable approach with a shift-share IV based on historical immigrant settlement patterns prior to Australia's 1996 immigration policy reform. To assess the credibility of our identification strategy, we follow Goldsmith-Pinkham et al. (2020) and report Rotemberg weights by country of origin. The identifying variation is concentrated in Asian countries such as India, China, the Philippines, Vietnam, and Malaysia, and the just-identified estimates for each of these countries are positive and similar in magnitude to the overall estimate.

2nd paragraph: summary (results) Our main results show that immigration raises employment and shifts the occupational composition of local labor markets. An increase

in the immigration share raises the employment share and lowers the share of the population not in the labor force, while it has no significant effect on the unemployment rate. The immigration shock also reshapes the occupational composition of employment: the share of high-wage occupations increases and the share of middle-wage occupations declines, consistent with a pattern of occupational upgrading. These findings are robust to leave-out IV specifications that exclude India and China, the two origin countries with the largest Rotemberg weights, from the instrument. The positive effect on employment is larger for men than for women, and the decline in the share not in the labor force is statistically significant for men. For both men and women, the share of high-wage occupations increases and the share of middle-wage occupations decreases, with a slightly larger increase in the high-wage share for women.

3rd paragraph: policy implication These findings carry important policy implications. The results indicate that immigration in Australia, largely driven by skill-based immigration from Asian origin countries, has been associated with higher employment and occupational upgrading in local labor markets rather than job displacement. At the same time, the decline in the share of middle-wage occupations suggests that the adjustment to immigration involves a reallocation of employment across occupations. Labor market policies that support workers in moving into higher-wage occupations, such as training and skill upgrading programs, may therefore complement immigration policy in areas with large immigrant inflows.

References

Australian Bureau of Statistics (August 2025b). Employee earnings, August 2025. ABS. https://www.abs.gov.au/statistics/labour/earnings-and-working-conditions/employee-earnings/latest-release.

Australian Bureau of Statistics (June 2025a). Australia's population by country of birth. ABS. https://www.abs.gov.au/statistics/people/population/australias-population-country-birth/latest-release.

Bartel, A. P. (1989). Where do the new us immigrants live? *Journal of labor economics*, 7(4):371–391.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of economic studies*, 89(1):181–213.

Card, D. (2001). Immigrant inflows, native outflows, and the local labor market impacts of higher immigration. *Journal of Labor Economics*, 19(1):22–64.

Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *The American Economic Review*, 110(8):2586–2624.

Hunt, J. and Gauthier-Loiselle, M. (2010). How much does immigration boost innovation? *American Economic Journal: Macroeconomics*, 2(2):31–56.

Jaeger, D. A., Ruist, J., and Stuhler, J. (2018). Shift-share instruments and the impact of immigration. Technical report, National Bureau of Economic Research.

Kim, J. S. and Wang, L. C. (2024). The differential effects of exchange rate fluctuations on local housing price growth: evidence from Australia. *Regional Studies*, 58(1):135–150.

Larsen, G. (2013). Family Migration to Australia. Technical report, Parliament of Australia.

Miller, P. W. (1999). Immigration policy and immigrant quality: The australian points system. *American Economic Review*, 89(2):192–197.

Peri, G. (2012). The effect of immigration on productivity: Evidence from us states. *Review of Economics and Statistics*, 94(1):348–358.

Wright, T. J. (2026). Replication of how much does immigration boost innovation?. *Economic Inquiry*, 64(1):8–21.

Figures

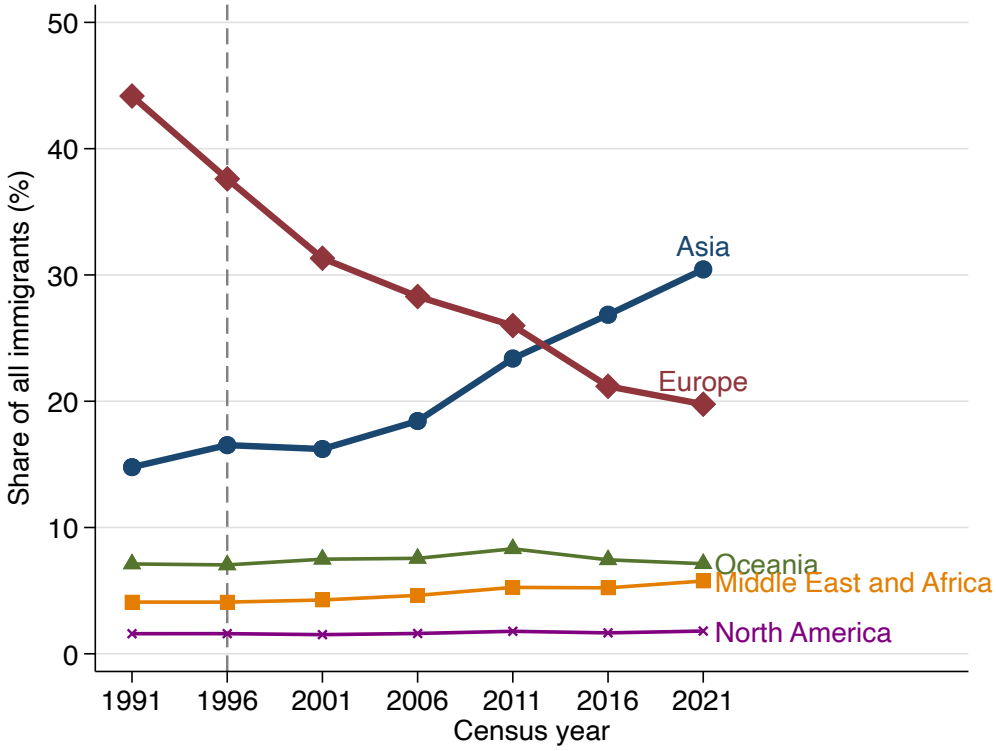


Figure 1: Immigrant population by birthplace, 1991 to 2021

Tables

Table 1: Summary statistics

Variables	Mean	St. Dev	Min	Max	N
Panel A. Immigration exposure					
Immigration share	0.277	0.227	0.000	2.557	2,480
Panel B. Labor force status					
Employed (share of population 15+)	0.563	0.077	0.183	0.837	2,480
Not in labor force (share of population 15+)	0.336	0.074	0.069	0.682	2,480
Unemployed (share of labor force)	0.056	0.027	0.000	0.380	2,480
Panel C. Occupational composition (share of total employed)					
High-wage	0.354	0.106	0.138	0.745	2,480
Middle-wage	0.424	0.085	0.109	0.727	2,480
Low-wage	0.202	0.049	0.067	0.501	2,480

Table 2: Rotemberg weights for employment share

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.026	-0.004	0.025
$\hat{\alpha}_k > 0$	1.026	0.047	0.975

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{Var}(z_k)$
$\hat{\alpha}_k$	1.000				
g_k	0.582	1.000			
$\hat{\beta}_k$	-0.127	-0.025	1.000		
$\hat{F}_k$	0.314	0.189	-0.353	1.000	
$\text{Var}(z_k)$	-0.171	-0.226	0.684	-0.437	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share (%)
India	0.281	3.3e+05	0.320	(0.208, 0.431)	1.480
China	0.166	3.5e+05	0.299	(0.143, 0.456)	1.891
Philippines	0.043	1.8e+05	0.406	(0.095, 0.716)	1.769
Vietnam	0.042	2.0e+05	0.444	(0.072, 0.816)	2.957
Malaysia	0.041	1.2e+05	0.256	(0.144, 0.368)	1.739

Panel D: Estimates of $\hat{\beta}_k$ for positive and negative weights

	α-weighted sum	Share of overall $\hat{\beta}$	Mean
$\hat{\alpha}_k \leq 0$	-0.023	-0.070	1.317
$\hat{\alpha}_k > 0$	0.353	1.070	0.998

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). The weights are computed at the origin country × year level and aggregated to the origin country level: $\hat{\alpha}_k = \sum_t \hat{\alpha}_{kt}$ and $\hat{\beta}_k = \sum_t (\hat{\alpha}_{kt}/\hat{\alpha}_k)\hat{\beta}_{kt}$. Panel A reports the sum, mean, and share of negative and positive Rotemberg weights $\hat{\alpha}_k$; Share is the share of the absolute sum of weights. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k, the national stock of immigrants born in country k, averaged over 2001–2021), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage Kleibergen-Paap F-statistics of each origin country's instrument ($\hat{F}_k$), and the variance of the 1991 origin country shares $z_{ik} = M_{ik,1991}/(M_{k,1991} \cdot Pop_{i,1991})$ across LGAs ($\text{Var}(z_k)$). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share is each origin country's share of all immigrants in Australia in 1991 (multiplied by 100). Panel D reports how $\hat{\beta}_k$ varies with positive and negative weights; the two α-weighted sums add up to the Bartik 2SLS estimate (0.330). The residual country group is excluded from Panel C but included in the estimation and in all other panels.

Table 3: Top 10 origin countries by Rotemberg weight: employment share

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	$\hat{F}_k$	Share (%)
India	0.281	3.3e+05	0.320	(0.208, 0.431)	25.60	1.480
China	0.166	3.5e+05	0.299	(0.143, 0.456)	9.95	1.891
Philippines	0.043	1.8e+05	0.406	(0.095, 0.716)	8.17	1.769
Vietnam	0.042	2.0e+05	0.444	(0.072, 0.816)	3.50	2.957
Malaysia	0.041	1.2e+05	0.256	(0.144, 0.368)	13.29	1.739
Sri Lanka	0.040	8.9e+04	0.249	(0.144, 0.355)	10.84	0.898
United Kingdom	0.024	1.1e+06	0.286	(0.170, 0.403)	30.23	26.807
South Africa	0.023	1.4e+05	0.306	(0.075, 0.537)	7.70	1.186
Thailand	0.021	5.0e+04	0.438	(0.244, 0.632)	12.80	0.331
South Korea	0.020	7.3e+04	0.198	(0.066, 0.331)	17.87	0.498

Notes: This table reports the ten origin countries with the largest Rotemberg weights, following Goldsmith-Pinkham et al. (2020). $\hat{\alpha}_k$ is the Rotemberg weight; g_k is the national stock of immigrants born in country k, averaged over 2001–2021; $\hat{\beta}_k$ is the just-identified IV estimate using the instrument of origin country k; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; $\hat{F}_k$ is the first-stage Kleibergen-Paap F-statistic; Share is each origin country's share of all immigrants in Australia in 1991 (multiplied by 100). The ten countries together account for 70.1% of the total weight. The residual country group is excluded from this table but included in the estimation.

Table 4: Immigration and labor force status, OLS and fixed effects estimates

	Share of population aged 15+		Share of labor force
	(1)	(2)	(3)
	Employed	Not in labor force	Unemployed
Panel A. Pooled OLS			
Immigration share	0.011	-0.019*	0.014***
	(0.013)	(0.011)	(0.004)
LGA fixed effects	No	No	No
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
Observations	2,480	2,480	2,480
Panel B. FE			
Immigration share	0.072***	-0.052***	-0.004
	(0.012)	(0.013)	(0.004)
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
Observations	2,480	2,480	2,480

Notes: Panel A reports pooled OLS estimates without LGA fixed effects; Panel B reports fixed-effects (FE) estimates with LGA fixed effects. FE-IV estimates are reported in Table 5. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Immigration and labor force status, IV estimates

	Share of population aged 15+		Share of labor force
	(1)	(2)	(3)
	Employed	Not in labor force	Unemployed
Panel A. All			
** Second stage*			
Immigration share	0.330***	-0.134***	-0.026
	(0.064)	(0.044)	(0.025)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	33.538	8.355	1.115
A-R *p*-value	0.000	0.004	0.292
Observations	2,480	2,480	2,480
Panel B. Men			
** Second stage*			
Immigration share	0.380***	-0.180***	-0.033
	(0.072)	(0.051)	(0.026)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	34.116	11.387	1.736
A-R *p*-value	0.000	0.001	0.188
Observations	2,480	2,480	2,480
Panel C. Women			
** Second stage*			
Immigration share	0.264***	-0.068*	-0.017
	(0.055)	(0.039)	(0.027)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	26.466	2.631	0.403
A-R *p*-value	0.000	0.105	0.526
Observations	2,480	2,480	2,480
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: The effect of immigration on occupational composition

	Share of total employed		
	(1)	(2)	(3)
	High-wage	Middle-wage	Low-wage
Panel A. All			
** Second stage*			
Immigration share	0.306***	-0.275***	-0.008
	(0.078)	(0.065)	(0.036)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.126)	(0.126)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	39.566	54.718	0.047
A-R *p*-value	0.000	0.000	0.829
Observations	2,480	2,480	2,480
Panel B. Men			
** Second stage*			
Immigration share	0.280***	-0.273***	0.015
	(0.075)	(0.068)	(0.035)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	36.377	47.612	0.188
A-R *p*-value	0.000	0.000	0.665
Observations	2,480	2,480	2,480
Panel C. Women			
** Second stage*			
Immigration share	0.350***	-0.269***	-0.055
	(0.086)	(0.061)	(0.040)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	36.709	50.265	2.151
A-R *p*-value	0.000	0.000	0.143
Observations	2,480	2,480	2,480
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Robustness checks with leave-one-out IV - labor force status

	Share of population aged 15+		Share of labor force
	(1)	(2)	(3)
	Employed	Not in labor force	Unemployed
Panel A. Leave-one-out SSIV (India)			
** Second stage*			
Immigration share	0.334***	-0.135**	-0.034
	(0.074)	(0.053)	(0.030)
** First stage*			
IV	0.863***	0.863***	0.863***
	(0.217)	(0.217)	(0.217)
C-D F-stat	184.651	184.651	184.651
K-P F-stat	15.796	15.796	15.796
A-R F-stat	20.268	5.077	1.472
A-R *p*-value	0.000	0.025	0.226
Observations	2,480	2,480	2,480
Panel B. Leave-one-out SSIV (China)			
** Second stage*			
Immigration share	0.336***	-0.144***	-0.024
	(0.065)	(0.047)	(0.026)
** First stage*			
IV	0.991***	0.991***	0.991***
	(0.203)	(0.203)	(0.203)
C-D F-stat	253.498	253.498	253.498
K-P F-stat	23.740	23.740	23.740
A-R F-stat	37.325	9.044	0.939
A-R *p*-value	0.000	0.003	0.333
Observations	2,480	2,480	2,480
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

Notes: FE-IV estimates using leave-out shift-share instruments that exclude immigrants from India (Panel A) or China (Panel B). C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Robustness checks with leave-one-out IV - occupational composition

	Share of total employed		
	(1)	(2)	(3)
	High-wage	Middle-wage	Low-wage
Panel A. Leave-one-out SSIV (India)			
** Second stage*			
Immigration share	0.324***	-0.306***	0.006
	(0.095)	(0.083)	(0.040)
** First stage*			
IV	0.863***	0.863***	0.863***
	(0.217)	(0.217)	(0.217)
C-D F-stat	184.651	184.651	184.651
K-P F-stat	15.796	15.796	15.796
A-R F-stat	29.513	38.086	0.024
A-R *p*-value	0.000	0.000	0.878
Observations	2,480	2,480	2,480
Panel B. Leave-one-out SSIV (China)			
** Second stage*			
Immigration share	0.320***	-0.279***	-0.016
	(0.078)	(0.064)	(0.037)
** First stage*			
IV	0.991***	0.991***	0.991***
	(0.203)	(0.203)	(0.203)
C-D F-stat	253.498	253.498	253.498
K-P F-stat	23.740	23.740	23.740
A-R F-stat	49.833	64.565	0.182
A-R *p*-value	0.000	0.000	0.670
Observations	2,480	2,480	2,480
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

Notes: FE-IV estimates using leave-out shift-share instruments that exclude immigrants from India (Panel A) or China (Panel B). Outcomes are employment in each occupational group as a share of total employed. C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix

Table A1: Rotemberg weights for not in labor force share

Panel A: Negative and positive weights					
	Sum	Mean	Share		
$\hat{\alpha}_k \leq 0$	-0.026	-0.004	0.025		
$\hat{\alpha}_k > 0$	1.026	0.047	0.975		
Panel B: Correlations					
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	Var(z_k)
$\hat{\alpha}_k$	1.000				
g_k	0.582	1.000			
$\hat{\beta}_k$	0.076	0.076	1.000		
$\hat{F}_k$	0.314	0.189	0.220	1.000	
Var(z_k)	-0.171	-0.226	-0.749	-0.437	1.000
Panel C: Top 5 origin countries					
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share (%)
India	0.281	3.3e+05	-0.132	(-0.193, -0.070)	1.480
China	0.166	3.5e+05	-0.084	(-0.164, -0.003)	1.891
Philippines	0.043	1.8e+05	-0.207	(-0.415, 0.000)	1.769
Vietnam	0.042	2.0e+05	-0.151	(-0.368, 0.066)	2.957
Malaysia	0.041	1.2e+05	-0.116	(-0.193, -0.039)	1.739
Panel D: Estimates of $\hat{\beta}_k$ for positive and negative weights					
	α-weighted sum	Share of overall $\hat{\beta}$	Mean		
$\hat{\alpha}_k \leq 0$	0.001	-0.005	0.136		
$\hat{\alpha}_k > 0$	-0.134	1.005	-0.828		

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). The weights are computed at the origin country × year level and aggregated to the origin country level: $\hat{\alpha}_k = \sum_t \hat{\alpha}_{kt}$ and $\hat{\beta}_k = \sum_t (\hat{\alpha}_{kt}/\hat{\alpha}_k)\hat{\beta}_{kt}$. Panel A reports the sum, mean, and share of negative and positive Rotemberg weights $\hat{\alpha}_k$; Share is the share of the absolute sum of weights. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k, the national stock of immigrants born in country k, averaged over 2001–2021), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage Kleibergen-Paap F-statistics of each origin country's instrument ($\hat{F}_k$), and the variance of the 1991 origin country shares $z_{ik} = M_{ik,1991}/(M_{k,1991} \cdot Pop_{i,1991})$ across LGAs (Var(z_k)). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share is each origin country's share of all immigrants in Australia in 1991 (multiplied by 100). Panel D reports how $\hat{\beta}_k$ varies with positive and negative weights; the two α-weighted sums add up to the Bartik 2SLS estimate (-0.134). The residual country group is excluded from Panel C but included in the estimation and in all other panels.

Table A2: Rotemberg weights for high-wage occupation share

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.026	-0.004	0.025
$\hat{\alpha}_k > 0$	1.026	0.047	0.975

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{Var}(z_k)$
$\hat{\alpha}_k$	1.000				
g_k	0.582	1.000			
$\hat{\beta}_k$	0.118	0.020	1.000		
$\hat{F}_k$	0.314	0.189	0.328	1.000	
$\text{Var}(z_k)$	-0.171	-0.226	-0.148	-0.437	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share (%)
India	0.281	3.3e+05	0.262	(0.146, 0.378)	1.480
China	0.166	3.5e+05	0.236	(0.048, 0.424)	1.891
Philippines	0.043	1.8e+05	0.526	(0.164, 0.887)	1.769
Vietnam	0.042	2.0e+05	0.417	(0.012, 0.822)	2.957
Malaysia	0.041	1.2e+05	0.181	(0.024, 0.338)	1.739

Panel D: Estimates of $\hat{\beta}_k$ for positive and negative weights

	α-weighted sum	Share of overall $\hat{\beta}$	Mean
$\hat{\alpha}_k \leq 0$	-0.009	-0.028	-0.283
$\hat{\alpha}_k > 0$	0.315	1.028	0.251

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). The weights are computed at the origin country × year level and aggregated to the origin country level: $\hat{\alpha}_k = \sum_t \hat{\alpha}_{kt}$ and $\hat{\beta}_k = \sum_t (\hat{\alpha}_{kt}/\hat{\alpha}_k)\hat{\beta}_{kt}$. Panel A reports the sum, mean, and share of negative and positive Rotemberg weights $\hat{\alpha}_k$; Share is the share of the absolute sum of weights. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k, the national stock of immigrants born in country k, averaged over 2001–2021), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage Kleibergen-Paap F-statistics of each origin country's instrument ($\hat{F}_k$), and the variance of the 1991 origin country shares $z_{ik} = M_{ik,1991}/(M_{k,1991} \cdot Pop_{i,1991})$ across LGAs ($\text{Var}(z_k)$). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share is each origin country's share of all immigrants in Australia in 1991 (multiplied by 100). Panel D reports how $\hat{\beta}_k$ varies with positive and negative weights; the two α-weighted sums add up to the Bartik 2SLS estimate (0.306). The residual country group is excluded from Panel C but included in the estimation and in all other panels.

Table A3: Rotemberg weights for middle-wage occupation share

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.026	-0.004	0.025
$\hat{\alpha}_k > 0$	1.026	0.047	0.975

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	Var(z_k)
$\hat{\alpha}_k$	1.000				
g_k	0.582	1.000			
$\hat{\beta}_k$	-0.119	0.048	1.000		
$\hat{F}_k$	0.314	0.189	-0.320	1.000	
Var(z_k)	-0.171	-0.226	-0.088	-0.437	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share (%)
India	0.281	3.3e+05	-0.194	(-0.283, -0.106)	1.480
China	0.166	3.5e+05	-0.253	(-0.424, -0.081)	1.891
Philippines	0.043	1.8e+05	-0.550	(-0.953, -0.147)	1.769
Vietnam	0.042	2.0e+05	-0.325	(-0.630, -0.020)	2.957
Malaysia	0.041	1.2e+05	-0.140	(-0.262, -0.018)	1.739

Panel D: Estimates of $\hat{\beta}_k$ for positive and negative weights

	α-weighted sum	Share of overall $\hat{\beta}$	Mean
$\hat{\alpha}_k \leq 0$	-0.002	0.008	1.151
$\hat{\alpha}_k > 0$	-0.272	0.992	-0.255

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). The weights are computed at the origin country × year level and aggregated to the origin country level: $\hat{\alpha}_k = \sum_t \hat{\alpha}_{kt}$ and $\hat{\beta}_k = \sum_t (\hat{\alpha}_{kt}/\hat{\alpha}_k)\hat{\beta}_{kt}$. Panel A reports the sum, mean, and share of negative and positive Rotemberg weights $\hat{\alpha}_k$; Share is the share of the absolute sum of weights. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k, the national stock of immigrants born in country k, averaged over 2001–2021), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage Kleibergen-Paap F-statistics of each origin country's instrument ($\hat{F}_k$), and the variance of the 1991 origin country shares $z_{ik} = M_{ik,1991}/(M_{k,1991} \cdot Pop_{i,1991})$ across LGAs (Var(z_k)). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share is each origin country's share of all immigrants in Australia in 1991 (multiplied by 100). Panel D reports how $\hat{\beta}_k$ varies with positive and negative weights; the two α-weighted sums add up to the Bartik 2SLS estimate (-0.275). The residual country group is excluded from Panel C but included in the estimation and in all other panels.

Table A4: Occupational composition, share of population aged 15+

	Share of population aged 15+		
	(1)	(2)	(3)
	High-wage	Middle-wage	Low-wage
Panel A. All			
** Second stage*			
Immigration share	0.316***	-0.034	0.054**
	(0.065)	(0.025)	(0.024)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	58.920	2.561	5.081
A-R *p*-value	0.000	0.110	0.025
Observations	2,480	2,480	2,480
Panel B. Men			
** Second stage*			
Immigration share	0.348***	-0.043	0.086***
	(0.071)	(0.031)	(0.030)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	65.363	2.875	8.515
A-R *p*-value	0.000	0.091	0.004
Observations	2,480	2,480	2,480
Panel C. Women			
** Second stage*			
Immigration share	0.284***	-0.032	0.015
	(0.059)	(0.023)	(0.021)
** First stage*			
IV	0.807***	0.807***	0.807***
	(0.167)	(0.167)	(0.167)
C-D F-stat	247.001	247.001	247.001
K-P F-stat	23.455	23.455	23.455
A-R F-stat	46.332	2.483	0.538
A-R *p*-value	0.000	0.116	0.464
Observations	2,480	2,480	2,480
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes

Notes: FE-IV estimates. Outcomes are employment in each occupational group as a share of the population aged 15 and above. C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. Control variables are linear time trends interacted with the population share aged 65 and above and the college graduate share, both measured five years before the start of the sample period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.